

## BEYOND THE TEXTBOOK · GUIDE 04

# Publish with GitHub Pages

Turn the files on your laptop into a public website with a free link.

|                                                                   |                                                                    |
|-------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>TIME NEEDED</b><br>15–20 minutes                               | <b>LEVEL</b><br>Beginner                                           |
| <b>YOU NEED</b><br>An email address and a finished project folder | <b>WORKS WITH</b><br>Guides 01, 02 and 03, or any HTML/CSS/JS site |

**WHAT YOU WILL BUILD**

A live website at an address like `https://your-username.github.io/my-portfolio/` that anyone can open on any device, free of cost.

## Before you start

- GitHub is where developers store and share code. **GitHub Pages** turns a code folder (a **repository**) into a website.
- Your project folder must have a file named exactly `index.html`. That is the page visitors see first.
- GitHub requires users to be at least 13 years old. Your repository and website will be **public**.

## Step 1 Create a GitHub account

1. Go to [github.com/signup](https://github.com/signup).
2. Enter your email, a strong password and a **username**. Choose it carefully: it becomes part of your website address. `asha-patil` looks more professional than `cooldude2008`.
3. Verify your email address.

**TIP**

Students can apply for free extras, including GitHub Copilot, through the GitHub Student Developer Pack: [education.github.com/pack](https://education.github.com/pack).

## Step 2 Create a repository

1. Click the **+** icon at the top right, then **New repository**, or go to [github.com/new](https://github.com/new).
2. **Repository name**: for example `my-portfolio` (lowercase, no spaces).
3. Select **Public**.
4. Tick **Add a README file**.
5. Click **Create repository**.

### GOOD TO KNOW

Special name: if you name the repository ``your-username.github.io`` (with your real username), the site is published at the main address `https://your-username.github.io/` with no folder name at the end. Good for your main portfolio.

## Step 3 Upload your files

1. In your new repository, click **Add file** → **Upload files**.
2. Open your project folder on your computer, select **all the files and folders inside it** (`index.html`, `style.css`, `images`, ...), and drag them onto the GitHub page.
3. Wait until every file appears in the list.
4. At the bottom, write a short message such as `Add portfolio files` and click **Commit changes**.

### BE CAREFUL

Upload the **contents** of the folder, not the folder itself. `index.html` must be visible on the repository's main page. If it sits inside another folder, your link will show a 404 error.

## Step 4 Switch on GitHub Pages

1. In the repository, click **Settings** (the gear tab at the top).
2. In the left sidebar, under **Code and automation**, click **Pages**.
3. Under **Build and deployment** → **Source**, choose **Deploy from a branch**.
4. Under **Branch**, choose **main** and the folder **/ (root)**, then click **Save**.

## Step 5 Open your live website

1. Wait one to two minutes. You can follow progress in the **Actions** tab (a green tick means done).
2. Refresh the **Settings** → **Pages** screen. At the top it says **"Your site is live at ..."** with your link.
3. Click **Visit site**. Open the same link on your phone too.

### Your website address

`https://your-username.github.io/my-portfolio/`

## Step 6 Update your website

Every time you commit a change, GitHub Pages rebuilds your site automatically within a minute or two.

- **Small edit:** open the file on GitHub, click the **pencil** icon, change the text, then **Commit changes**.
- **Many changes:** edit the files on your computer, then use **Add file** → **Upload files** again. Files with the same name are replaced.

**TIP**

If you still see the old version, press **Ctrl + F5** to reload without the browser's saved copy.

## Step 7 Publishing the whole starter repository

If you copy the full workshop starter repository to your account and switch on Pages, each example gets its own address:

| Page                   | Address                                                    |
|------------------------|------------------------------------------------------------|
| Home page              | <code>https://your-username.github.io/repo-name/</code>    |
| Portfolio              | <code>.../repo-name/examples/01-portfolio/</code>          |
| Formspree contact form | <code>.../repo-name/examples/02-formspree-contact/</code>  |
| Supabase guestbook     | <code>.../repo-name/examples/03-supabase-guestbook/</code> |

## Step 8 Check quality and share

1. Test your live link with [PageSpeed Insights](#). It scores speed, accessibility, best practices and SEO, and lists what to fix.
2. Test every link, button and form on the live site.
3. Add the link to your CV, LinkedIn profile, college applications and the repository's **About** box (gear icon on the repository's main page).

## Optional: use Git commands

As you grow, you will use Git from VS Code's terminal instead of uploading files in the browser. Install Git from [git-scm.com](https://git-scm.com), then:

**Terminal**

```
git clone https://github.com/your-username/my-portfolio.git # copy the repo to your
computer
cd my-portfolio
# ...edit your files...
git add . # choose all changed files
git commit -m "Update project section" # save a version with a message
git push # send it to GitHub (site updates
automatically)
```

Learn more in the free guide at [docs.github.com/get-started](https://docs.github.com/get-started).

### ASK AN AI HELPER (Claude, Gemini, Copilot or Antigravity)

"My GitHub Pages site shows a 404 error. Here is a list of my repository files. What is wrong?"

"My CSS works on my laptop but not on GitHub Pages. Here is my link tag and file list."

"Explain git add, git commit and git push in simple words with an example."

"Write a short, professional README.md for my portfolio repository."

## Troubleshooting

| Problem                                       | Likely cause and fix                                                                                                                                       |
|-----------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 404 "There isn't a GitHub Pages site here"    | Wait two minutes. Check that <code>index.html</code> (all lowercase) is at the top level of the repository and that Pages is set to <b>main / (root)</b> . |
| Page loads but has no styling                 | The CSS path is wrong. Use a relative path like <code>href="style.css"</code> , not <code>C:\Users\....</code> . Check upper and lower case.               |
| Images missing online but fine on your laptop | Case mismatch: <code>Photo.JPG</code> vs <code>photo.jpg</code> . GitHub is case-sensitive; Windows is not.                                                |
| Old version still showing                     | Press <code>Ctrl + F5</code> , or wait for the green tick in the <b>Actions</b> tab.                                                                       |
| No Pages option for a private repository      | On a free account, GitHub Pages needs a <b>public</b> repository.                                                                                          |

## Safety

- Everything in a public repository can be seen by anyone, including old versions.
- Never upload passwords, secret keys (`sb_secret_...`, `service_role`), or private documents. If you upload one by mistake, deleting the file is not enough: change (rotate) the key immediately in that service's dashboard.
- The Formspree form ID and the Supabase **publishable** key are designed to be public.

## Final checklist

- ☐ My repository is public and `index.html` is at the top level.
- ☐ Pages is set to **Deploy from a branch** → **main** → **/ (root)**.
- ☐ My live link opens on both laptop and phone.
- ☐ Links, images, forms and the guestbook work on the live site.
- ☐ No passwords or secret keys are in my repository.

## About the Instructor

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